

Day 57 Research Log — Flat Memory Pivot (May 25, 2026)

A brain you can fit in 233 MB that absorbs frontier-scale text and memory never grows. Not a model. An organism.

TL;DR

Across 13 packs (110–126) we replaced the unbounded `{word: HV}` dict lexicon at the core of Ikigai with a **fixed-size Sparse Distributed Memory** holding every channel. The trained organism now lives in **192 MB of substrate that never grows**, loads at **233 MB inference RAM**, and answers similarity / IS-A / sensory / arithmetic / generation queries through one substrate. The dict is dead.

The architecture is the literal cerebellar model (Marr–Albus–Kanerva 1988) augmented with **computed-identity keys** (zero per-word storage), **VSA role-binding** (one substrate, many channels), **read-time mean-removal** (sensory adaptation), and **traffic-class bank separation** (sparse signals don't drown in dense ones). Five-way integration unbuilt elsewhere.

The Bet

Frontier LLMs grow parameters with data. Brains do not — adult human cortex is ~86B neurons fixed. Lifetime experience lives in synaptic weights, not in a growing dictionary. The architectural promise: **frontier-data-worth of knowledge in <1 GB, biologically grounded, online-learning, no backprop, CPU-only**.

The dict lexicon (Pack 113 audit: 98% of organism RAM) is the disease. Compressing it (d=512, float16) just slows the bleed. The cure: replace the dict entirely with a constant-RAM substrate where memories superpose.

The Migration (Packs 110–126)

Pack	Goal	Result
110	Organism <code>save / load</code>	12/12. 243 MB pickle, 0.7 s round-trip
111	10K diversified corpus	TS + wikitext-103 + math; Channel 4 noise fixes
112	Persona corpus	10/10 after fixing <code>pgmw._normalize</code> (L2 vs component-wise) bug

Pack	Goal	Result
113	RAM audit	Lexicon = 98% of RAM. d=2048 dict will hit 1.6 GB at 1M exposures
114	Flat memory POC	8/8. Vocab 0 → 14,429, substrate +0 bytes ever
115	Mean-removal sharpening	sep 0.026 → 0.371 (14×). IDF write-gating HURTS — read-time adaptation > write-time gating
116	Channel-1 migration + speed	Flat beats dict sep 0.370 > 0.152. Batched activation 1.5×
117	Multi-role unifier	One substrate, all channels via VSA role-binding. Traffic-class bank separation discovered
118	Wired into organism	IS-A 10/10 after <code>assert_isa(n=50)</code> reinforcement. Math word problems answered
119	Capacity stress + consolidation	Flat survives 30K stories. Full-renorm consolidation is wrong policy (kills mean-removal signal)
120	Generation off dict	Bigram-equivalent via flat. <code>word_weight=4</code> key mix kills <code>little/little</code> collisions
121	Verb rotor (Channel 2) on flat	8/8. Math 12/12 within 5% relative
122	Dict containers emptied	<code>being.lexicon = 0, grammar._bigram = 0</code> . Boy does math word problems
123	RAM at 60K scale	Substrate FIXED 192 MB; pyarrow/numpy bloat exposed as training-time artifact
124	Scaffolding gutted	Inference RAM = 233 MB. Substrate 192 of that
125	Past crossover (100K)	Vocab 19K, flat wins 1.58× vs original d=2048 dict
126	Skill gallery	chain 5/5 (cat → mammal → warm). isa 6/6, sensory 6/6, math 7/10

Architecture

ComputedKey(word)

= renorm($\sum$ phasor(char_trigrams(word)) + 4 · phasor(WORD:word))
Zero stored bytes per word. Char-trigrams give OOV generalization;
whole-word phasor mix ensures distinct words → orthogonal keys.

VSASDM (per traffic class):

Hconj $\in \mathbb{C}^{M \times d}$ (random hard-location addresses, regen from seed)
C $\in \mathbb{C}^{M \times d}$ (learned counter bank)
write(addr, data): top-k activation → C[idx] += data
read(addr): sum activated rows → renorm
Cost: $2 \cdot M \cdot d \cdot 8$ bytes FIXED. Dense bank M=16384, sparse M=8192.

MultiRoleMemory (the organism's substrate):

Address per (word, role) = key(word) $\odot$ ROLE_phasor
Different roles → orthogonal addresses → different hard locations.
Roles: cooccur, next (dense bank);
isa, sensory, property, verb (sparse bank).

Adaptive recall:

meaning(w) = read(addr) - $\sum_{v \in \text{top-r}} (v, \text{read}) \cdot v$
Common-direction removal (Arora all-but-the-top). Scale-invariant.

Verb rotor encoding:

c_est = (n_after - n_before) / modifier
hv = exp(i · c_est · ω · q_axis)
Superposed observations → median(angle/ ω /q_axis) on recall = mean c.

Generation:

expose_transitions writes (prev → curr) under 'next' role.
next_word_candidates(prev) = vectorized cleanup of read(addr) against vocab.
flat_generate: Markov walk over flat-memory transitions.

The Numbers

Substrate:	192 MB FIXED forever, any data volume
Pack 122 (5K TS+5K math+facts) checkpoint:	161 MB pickle
Inference RAM (Pack 124):	233 MB total RSS
Pack 125 100K stories vocab:	19,411
vs original d=2048 dict at 19K vocab:	303 MB (flat wins 1.58×)

The 3 GB observed during Pack 123 training was pyarrow + numpy heap fragmentation — engineering bloat around the substrate, not the substrate. At deployment (load checkpoint, query) RAM is 233 MB.

What the Boy Does (Pack 126 highlights)

```
Similarity:  cat ~ mouse(0.68), bird(0.59), dog(0.53)
             big ~ small(0.64), bed(0.48), yellow(0.44)
```

```
IS-A (6/6):  dog → mammal,  oak → tree,  apple → fruit,
             rose → flower, car → vehicle
```

```
Sensory (6/6): red → R_VISUAL_RED,  hot → R_TEMP_HOT,
               big → R_SIZE_BIG,    cold → R_TEMP_COLD
```

Math word problems (7/10 within 8% rel-tol):

"Bob had 50 dollars. He spent 18. How many remain?" → 32.0 (truth 32)

"Mary had 12 apples. She ate 4." → 8.2 (truth 8)

"Pat had 40 books. He got 15 more." → 55.0 (truth 55)

Cross-channel composition (5/5) — chain through ONE substrate:

```
cat   → isa → mammal → property → warm
oak   → isa → tree   → property → tall
car   → isa → vehicle → property → fast
apple → isa → fruit  → property → sweet
```

The chain is the structural payoff. Two role queries through one memory compose `property(isa(cat)) = warm`.
This is the cross-channel binding that the separate-channel dict organism never had.

What Is Novel

Each piece exists in literature; the **integration** does not.

1. **Computed (0-storage) keys + whole-word phasor mix.** `word_weight=4` discovery: char-trigram-only keys collapse lexically similar words (`cos little/litttle = 0.85`). Mixing a whole-word random phasor drops that to 0.09 while preserving OOV trigram signal.
2. **Multi-role SDM with traffic-class banks.** VSA role-binding makes addresses orthogonal; the discovery: a *shared* counter bank still lets a dense channel (cooccur, ~1400 writes/loc) swamp a sparse one (one isa fact, ~1 write/loc) at shared locations. Fix: separate banks per traffic class. This is not in any paper.

3. **Mean-removal at SDM readout.** Arora 2017's all-but-the-top for static embeddings. Applying it to noisy SDM recall: sep 0.026 $\rightarrow$ 0.371 (14 $\times$), unrelated baseline +0.88 $\rightarrow$ -0.06. First application to a content-addressable memory.
4. **Write-count = synaptic strength for cleanup.** Repeated `assert_isa(n=50)` lays down 50 $\times$ magnitude; cleanup picks highest-magnitude target. No separate count dict. Biology-aligned, implementationally elegant.
5. **Verb-as-rotor flat encoding.** Scalar c encoded as `phase = c *  $\omega$  * q_axis`, superposed across observations, decoded by median angle. Running-average HV arithmetic in a content-addressable memory. New.
6. **Phasor HDC throughout.** Component-wise unit phasors in $\mathbb{C}^d$ for *both* Hebbian drift *and* VSA binding *and* verb rotors. One representation for symbolic-and-grounded. Most VSA work uses bipolar/binary.
7. **The integration.** Computed keys + Hebbian-written fixed SDM + read-time mean-removal + VSA role-binding + complete online-learning language organism. Five-way. Unbuilt elsewhere.

Differences from Existing Systems

	Transformers	RAG	Vector DBs	Kanerva SDM (1988)	Ikigai
RAM scaling	grows w/ params	dict + LLM grows	grows w/ corpus	substrate fixed	substrate fixed
Training	batch backprop GPU	requires LLM	n/a	per-write Hebbian	online Hebbian CPU
New facts	retrain (\$M)	re-embed	re-index	one write	one write
Forgetting	catastrophic	none (lookup)	none	graceful crosstalk	graceful crosstalk
Identity storage	embedding table	embedding table	id $\rightarrow$ vec	per-key in bank	computed, 0 bytes
Multi-channel	single model	RAG = lookup only	n/a	one role	N roles, one substrate
Bio mapping	loose	none	none	cerebellum	cerebellum + Hebbian + adaptation

Open Problems

1. **Channel-2 structural crosstalk (~5%).** At $M_{rel}=8192$, $k=64$, 8-verb writes share enough hard locations to bias decoded coefficients by 5–7%. Bigger M_{rel} reduces it; principled fix is to use per-verb partitioned addresses or higher-rank verb representation.
2. **Generation coherence beyond 6–8 tokens.** Flat-memory bigrams + Markov walk drifts off-topic. Trigrams/4-grams as additional `next2/next3` roles could extend coherence with constant-RAM cost.

3. **Consolidation policy.** Full L2 renorm kills the magnitude differentials mean-removal needs. Better policies (cap-not-renorm, periodic decay $C \leftarrow C \cdot 0.95$) untested. Pack 119 confirmed mean-removal alone suffices at $\leq 30K$ scale.
 4. **Capacity wall past 100K stories.** Pack 125 vocab 19K; TinyStories saturates near 30K. Past crossover for $d=512$ dict equivalent needs higher-diversity corpora (wikitext, encyclopedia).
 5. **Training-time RAM** (3 GB at 60K) vs **inference RAM** (233 MB). Inference is the deliverable; training bloat is HuggingFace pyarrow + numpy heap fragmentation — solvable with explicit buffer reuse + raw text I/O.
 6. **Dict scaffolding still allocated in non-flat mode.** `flat_only=True` is opt-in. Default organism still builds the full cognition stack. Deprecation path: make `flat_only=True` the default once all callers ported.
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What's Next (Day 58+)

- **Scale corpora.** Wikipedia + curated factual sources; push vocab past 50K to clear the $d=512$ dict crossover empirically.
 - **Real benchmarks.** Run the no-forgetting benchmark (Day 53) on the flat-only organism. 5 novel patterns + 20 distractor patterns + recall test.
 - **Inference engine.** Strip HF entirely; load checkpoint + serve queries via a tiny CLI. Measure cold-start, throughput, persistent RAM.
 - **Multi-organism merge.** Two flat substrates with the same seed $\rightarrow$ counter banks SUM = federated learning by addition. Test cross-organism knowledge merge.
 - **Vision channel.** Pixel encoder $\rightarrow$ HV $\rightarrow$ bound to sensory anchors via existing role-binding. The substrate is modality-agnostic.
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Closing

The dict was the disease. The substrate is the cure. **192 MB of fixed counters carry every channel of an online-learning language organism that doesn't forget, doesn't need GPUs, and never grows.** Frontier-data knowledge in <1 GB is real at inference. Five-way integration nobody built before.

Not a model. An organism.

— Mura ALife Labs